Chapter 17

Uncertainty

Learning Objectives

- 17.1 Assign simple probability distribution to random variables and compute expected costs and benefits.
- 17.2 Explain the use of price discrimination to model pricing uncertainty.
- 17.3 Examine the use of selection bias and randomized experiments to resolve the uncertainty around business decisions.
- 17.4 Analyze situations that minimize expected error costs.
- 17.5 Assess situations to get accurate estimates of uncertainty.
- 17.6 Design institutions to deal with unforeseen contingencies.

Chapter 17 – Main Points

- When you're uncertain about the costs or benefits of a decision, assign a simple probability distribution to the variable and compute expected costs and benefits.
- When customers have unknown values, you face a familiar trade-off: Price high and sell only to high-value customers, or price low and sell to all customers.
- If you can identify high-value and low-value customers, you can price discriminate and avoid the trade-off. To avoid being discriminated against, high-value customers will try to mimic the behavior and appearance of low-value customers.
- Difference-in-difference estimators are a good way to gather information about the benefits and costs of a decision. The first difference is before versus after the decision or event. The second difference is the difference between a control and an experimental group.
- If you are facing a decision in which one of your alternatives would work well in one state of the world, and you are uncertain about which state of the world you are in, think about how to minimize expected error costs.

Warm-Up

- Have you ever made a big decision (like choosing a major, moving cities, or buying a car) and later realized you didn’t have all the information? What did you wish you had known?

Uncertainty

- Uncertainty occurs because of missing information
  - Few things are certain
  - We often classify risk and uncertainty as the same. However, there are some key distinctions
    - Risk is quantifiable
    - Uncertainty is when we don’t know the possible outcomes or their probabilities.
    - We have to gather more information to reduce the degree of uncertainty and turn it into risk
- To handle uncertainty, random variables (variables that take different values) are used
- These random variables are then assigned probabilities. The probabilities act as a likely outcome that the assigned value occurs
- The summation of these random variable outcomes is known as the expected value (the average value or outcome)

Uncertainty

- Consider the expected value: the sum of n events that occur with n probabilities that total 1.

$$E(X) = \sum_{i=1}^{n} p_1 * x_1 + p_2 * x_2 + \cdots + p_n * x_n,$$

$$s.t. 0 < p < 1 \;\&\; \sum p_i = 1$$

Example: You have a fair coin. You win $5.00 if the coin lands on heads, and you lose $3.00 if the coin lands on tails.

What is the expected value of the play?

$$E(X) = 0.5(5.00) + 0.5(-3.00)$$
$$E(X) = \$1.00$$

Example

You have a six-sided fair die. You win $2, $4, $6, $8, $10, and $12 if you roll a 1, 2, 3, 4, 5, 6 respectively. What is the expected value of the roll?

(1/6)*$2 + (1/6)*$4 + (1/6)*$6 + (1/6)*$8 + (1/6)*$10 + (1/6)*$12 = $7

What if you lose by rolling less than a four and win by rolling more than a three?

(1/6)*$12 + (1/6)*$10 + (1/6)*$8 - (1/6)*$6 - (1/6)*$4 - (1/6)*$2 = $3

Discussion Activity 1

- Suppose your startup can invest in a risky but potentially high-return technology. How would expected value help, and what might it miss?
- Think about how much variance there is. Is the upside worth the downside risk?

How Uncertainty Helps You

- Since you know some of the likely outcomes (assigned probabilities) you can make an educated guess as to the best outcome.
- To use uncertainty place probabilities on the success and failure of a project
- If you think a project will have a 60% probability of being successful then that project has 40% probability of failing.
- For example, reconsider the coin toss. All else equal, there is a 50% probability that you will win and a 50% probability you will lose.
  - If the coin lands on heads you win $1. If the coin lands on tail you lose $1.
  - The expected value of the coin toss is: 0.5($1) + 0.5(-$1) = $0.

- The next two questions that we should ask are:
  - 1) How do we assign probabilities?
  - 2) When should we undertake the project?

Answering Our Questions

1) Probabilities are determined in a few ways:

- Based on experience
- Estimate probabilities based on market data
- Run simulations
- The key takeaway is that no matter how well we think we know the probability of a particular outcome, it is best to run different scenarios (simulate a model) to see the variance in the expected values.

2) We undertake a project when the expected profit is greater than the expected loss.

- Be careful though. You may have estimated the wrong probabilities.
- It never hurts to gather more information.

Uncertainty Estimates Through Method Choice

Key Points:

- Not all forecasting methods are equal in reliability.
- Choosing how to estimate is just as important as what to estimate.

Method	Pros	Cons
Historical Data	Fast, cheap	May be outdated or misleading
Customer Surveys	Reflect preferences	Prone to bias, hypothetical
A/B Testing	Accurate, real behavior	Slower, costlier, sample limits
Market Simulation	High control, scenario planning	Requires assumptions, complexity

Scenario Example: Estimating demand for a new coffee blend.

- Historical data ≠ useful if it's a new flavor.
- A test in 3 stores for 4 weeks gives much better uncertainty bounds.

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Uncertainty Problem

- You have $50,000 that can be invested into project 1 or 2. Based on prior knowledge, you estimate project 1 will succeed with 0.60 probability in which case it triples in value.

- If project 1 fails, its scrap value is worth $25,000. You've estimated project 2 to succeed with a probability of 0.70, and will return 2.5 times its original value if it succeeds and $30,000 if it doesn't.

- In which project do you invest?

Uncertainty Problem cont.

- Project 1: $50,000 x 3 = $150,000 if successful.
- Project 2: $50,000 x 2.5 = $125,000 if successful.

- 60% chance of success in project 1: 0.6*150,000 = 90,000
- 70% chance of success in project 2: 0.7*125,000 = 87,500

- 40% chance of recouping scrap in project 1: 0.4*25,000 = 10,000
- 30% chance of recouping scrap in project 2: 0.3*30,000 = 9,000

- Project 1 nets: 90,000 + 10,000 = $100,000
- Project 2 nets: 87,500 + 9,000 = $96,500

Risk-Adjusted Expected Value

- Not all $100 expected profits are created equal—variance matters.
- Expected value = average outcome over many trials.
- But decisions must also consider **variance** (spread of possible outcomes).
- **Higher variance = more risk**, even if expected value is the same.

Example:

- Project A: 50% chance of $0, 50% chance of $200 → E[V] = $100
- Project B: 100% chance of $100 → E[V] = $100
  - But most risk-averse managers prefer Project B.

- Use expected value for *direction but* use variance to assess *risk appetite*.

Pricing Consumers out of the Market

- As discussed with demand, consumers aren't all the same. Some consumers value a product differently than other consumers.
- As a firm, it may be difficult to distinguish between these types of consumers so pricing becomes difficult.
- Suppose we separate our consumers into two types (though more could exist): high value and low value.
  - High value consumers will pay the high and low price for the good
  - Low value consumers will only pay the low price.
- If we price incorrectly, we run the chance of losing customers and profit.

Pricing Consumers out of the Market Example

- Suppose that half of our customers would pay $20 for our good and the other half would only pay $14. If the good has a marginal cost of $8 and we price the good at $20, what is the expected profit? What if you priced the good at $14?

$$E(\pi 1) = 0.5 * (\$20 - \$8) + 0.5 * (0) = 0.5 * (\$12) = \$6$$

$$E(\pi 2) = \$14 - \$8 = \$6$$

Discussion Activity 2

- Why would a company prefer to price-discriminate rather than set one flat price for everyone?
- Consider how pricing at one level may exclude some consumers while leaving money on the table with others.

Experiments

- We can utilize experiments to obtain more information, and allow us to make better decisions in the future.
  - These also help you analyze the current decision’s benefits and costs.
- These experiments entail having a control group and a treatment group. The control group does not experience the change being made, whereas the treatment group does.
- When the experiment is done, we compare the changes between the control and treatment group. These changes are called a **difference-in-difference** estimate.
  - That is, these measure the difference between groups and the change in policy/decision

Experiments

- When designing experiments be careful that you have a representative control group but also be wary of leakages.
  - Leakages occur if your control and treatment group are nearly identical or have a proximity.
    - E.g., You are selling a 2-door hatchback car and are considering price discounts. If you worry about leakages then using the 4-door version of the car would be a poor choice as they are very close to one another, and consumers are likely to purchase one or the other.
    - What would be a better choice to hedge against leakages? A truck?
  - Representativeness is how well your treatment group is reflected by your control group.
    - Consider the same car. If you want to have a better representative in your control group, then the 4-door version may be the better choice.
- In the leakage case, we want to move away from the treatment group if we are concerned too many sales go to the control group
- In the representative case, we want to get closer to the treatment group if we want to be sure our decision is a good one.

Avoiding Pitfalls in Business Experiments

Key Points:

- **Selection bias**: Non-random groups may differ in unseen ways.
- Contamination/leakage: Info or incentives from treatment spill into control group.
- Sample size too small: Can’t detect meaningful effects → false negatives.
- Ignoring external validity: Results may not generalize beyond test setting.
- Quick Checklist:
  - Random assignment
  - Equal treatment aside from variable tested
  - Proper controls for timing, seasonality, and spillovers
  - Pre-registration or protocol planning

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Difference-in-Difference

- D-I-D estimates are interesting in that they attempt to capture group effects/factors and time-varying effects/factors.
- The before and after of the treatment group may be labeled something like A and B.
  - This controls for factors that are constant to the group.
  - Ex: we targeted a particular age group and let others vary
- The before and after of the control group measures the time varying component of the experiment. We can label these C and D for before and after.
- In essence, we take two differences from the treatment and control groups and subtract them to find the effect of the experiment.

D-I-D estimate = (B – A) – (D – C)

Example

- You raise your product price by $20 in market 1 but leave it unchanged in market 2. Sales in market 1 fall from 900 units per week to 800 while sales in market 2 rise from 750 to 825. The difference-in-difference estimate of the effect of the price change is:

- Treatment group has their price changed
- Control group has no change in price

	Raise Price	**Keep Price**
Raise Price (T)	800	900
Keep Price (C)	750	825

- First difference minus second difference = difference-in-difference
- (800 – 900) – (750 – 825) = -25

Discussion Activity 3

- If you're testing a new app feature, how would you design an experiment to avoid bias and leakage?
- Think about your control group, location choice, sample size, and timing.

Selection Bias

- One problem that was in the background of the experiment was selection bias.
  - Occurs when the treatment group differs systematically from the control group.
- Selection bias adds some to the measured outcome

$$Observed\ effect = Treatment\ Effect + Selection\ Bias$$

- The problem with selection bias is that it overstates the importance of the observed effect.
- If we don't control for it, then our estimates are greater than what the causal effect should be.
- The simplest way to address selection bias is to conduct a randomized experiment
- When a randomized experiment can't be conducted, then we have to rely on statistical techniques or other data analysis to parse out the effect.

Expected Error Cost

- Many times we want to choose the best outcome, but uncertainty clouds our decisions
- In many instances, it's best to minimize your cost than maximize your profit. This is called minimizing expected error cost
- Given this uncertainty, it is possible to have a Type I Error or a Type II Error when making decisions
  - Type I Error – a false positive. (You thought something was there when it wasn't)
  - Type II Error – a false negative. (Something was there and you didn't think it was)

- A Type I Error occurs because you lost out on a profitable venture when you thought your original hypothesis of a non-profitable outcome was true.
- A Type II Error occurs because you incurred more costs when you thought your original hypothesis of a profitable outcome was true.

Expected Error Cost

- Deciding which decision to make can be best analyzed with the following:

$$\bar{p} = \frac{Error\ Cost\ I}{Error\ Cost\ I + Error\ Cost\ II}$$

- We want to minimize the expected cost of our decision, so we compare the estimated probability above to the expected probability of success.
- If $\bar{p} > p$ then we don't proceed with the project.
- If $\bar{p} < p$ then we do proceed with the project.
- The above provides the best chances for success.
- Note, that if you approve a project and it is a failure, then those results are readily known. If you don't approve a project, then the results aren't known because we never observed what could have happened.

Expected Error Costs

Decision Matrix:

	True Good Idea	True Bad Idea
Invest	Success	Type I error
Don't Invest	Type II error	Avoided loss

Example: A competitor releases a new product. Should you launch a response?

- Type I: Waste resources on a weak launch.
- Type II: Lose market share by doing nothing.

Expected Error Cost Example

- A professor wants to buy a new tablet for school use. The tablet costs \$400. He knows from past experience that the school will cover half the cost. If the school doesn't cover half the cost, he pays the full \$400. Suppose, he's willing to pay at most \$300. At what probability of expected success should he proceed?

- $E(C) = p \times 200 + (1 - p) \times 400$
- $E(C) \leq 300$
- $200p + 400 - 400p \leq 300$
- $-200p \leq -100$
- $p = 0.5$

- Interpretation: if, for example, based on historical data, he know that 4/8 purchases had been approved, then he would proceed with the purchase.

- If he knows the school will cover half the cost 75% of the time, should he proceed?

Activity - Investment Decision Under Risk

Scenario:

- A company is evaluating two projects:
  - Project A:
    - 40% chance of earning $120,000
    - 60% chance of earning $40,000
  - Project B:
    - 100% chance of earning $75,000

Questions:

1. What is the expected value of each project?
2. Which project is preferred by a risk-neutral manager?
3. What might a risk-averse manager prefer?
4. What if the company had a high-risk appetite?

Activity - Investment Decision Under Risk

- Project A:
  - 0.4(120,000) + 0.6(40,000) = 48,000 + 24,000 = $72,000
- Project B:
  - E(B) = 1.0(75,000) = $75,000

- A risk-neutral manager prefers Project B for the higher expected value.
- A risk-averse manager definitely prefers Project B for the lower risk and higher expected return.
- A risk-loving company may choose Project A if the upside ($120K) aligns with aggressive goals.

Designing for Uncertainty

Smart organizations plan for uncertainty by building in flexibility.

Design Tools:

- **Option-based contracts**: Allows deferral or cancellation with minimal penalty.
- **Real-time feedback systems**: Constantly update forecasts with incoming data.
- **Adaptive compensation schemes**: Incentives adjust as business conditions shift.
- **Scenario planning**: Predefined responses to likely disruptions (e.g., supply chain risk).

Example:

A supplier contract includes:

- Flex pricing based on input costs
- Exit clause if demand drops >20%
- Penalty waiver for force majeure events

Avoiding Bias in Business Experiments

- Experiment design checklist
- Random assignment
- Comparable control and treatment groups
- Control for time/season effects
- Watch for contamination/leakage
- Ensure external validity

Case Study: New Coke vs. Old Coke

- Coca-Cola spent three years testing a new formula of their popular Coca-Cola soft drink.
- Coca-Cola spent more than $4 million in the development of New Coke and even tested it out on over 200,000 consumers. The results were largely positive, so Coca-Cola decided to go ahead and introduce New Coke.
- New Coke came out on April 23, 1985.
- The public had an immediate backlash (some were so fearful that they stocked up their basements with old Coke). By July 11, 1985 and millions of advertising dollars later, Old Coke (or Coca-Cola Classic) replaced New Coke.

- Type II Error: Data suggested success, but reality showed backlash.
- Ignored external factors (nostalgia, brand attachment) not captured in tests.
- Could have reduced risk by simulating market reaction or using experimental test markets.

How could Coke have designed a better test? What did they miss about uncertainty?

Uncertainty vs. Risk – Summary & Tools

	Risk	**Uncertainty**
Definition	You know the possible outcomes and their probabilities	You don’t know the outcomes, probabilities, or both
Example	A coin flip, insurance payout, factory downtime	New product launch success, market reaction, pandemics
Best Tool	Expected value Variance/standard deviation Decision trees	Experiments Pilot tests Forecast ranges Contracts with contingencies
Decision Focus	Choose the best-known option	Reduce what you don’t know before choosing
Typical Errors	Miscalculating risk → wrong valuation	Acting too soon or too late due to missing information
Business Example	Insurance pricing, credit scoring	Coke’s new formula, launching in a new market

Key Takeaways

- Uncertainty = missing information; convert it into risk using probabilities and models.
- Use expected value to compare options but remember to consider variance (risk).
- Simulations and scenario analysis help visualize uncertainty and prepare for surprises.
- Price discrimination lets firms respond to demand uncertainty by segmenting customers.
- Experiments, especially difference-in-difference, help reduce uncertainty and test decisions.
- Watch out for selection bias, leakage, and poor sample size in experiments.
- Decisions under uncertainty should minimize expected error costs, not just maximize profit.
- Smart institutions plan for uncertainty using adaptive contracts, real-time feedback, and flexible systems.

Knowledge Check

1. What’s the difference between risk and uncertainty?
2. What does expected value measure?
3. How does variance affect decision-making even if expected value is the same?
4. What is a Type I vs. Type II error?
5. What does a difference-in-difference estimate control for?

Knowledge Check Answers

1. Risk is measurable with known probabilities; uncertainty involves unknowns.
2. The weighted average of all possible outcomes.
3. Higher variance means higher risk; some managers may avoid high-variance outcomes.
4. Type I: false positive; Type II: false negative.
5. Controls for both group-specific and time-specific changes.

Consultant’s Challenge: Deciding Under Uncertainty

Scenario:

- You’re consulting for a startup launching a new wearable health device. They’re debating two marketing strategies, but outcomes are uncertain, and they don’t know how to choose.

Your Task:

1. How would you help them think in terms of expected value to compare the two strategies?
2. What uncertainty should they reduce and what tools (like experiments or market segmentation) would help them do it?